Hybrid Cognitive-Fluid Dynamics for Language Production: A Wasserstein Gradient Flow Perspective

L'Electron Rare
contact@saillant.cc

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Abstract

We introduce a family of reaction-diffusion models for the four-species Levelt–Baddeley account of language production, formulated as a Wasserstein gradient flow on the space of activation distributions. Using a multi-scale loop that couples Langevin particle dynamics with an implicit JKO step, we show that (i) the flow is numerically tractable at scale on commodity Apple Silicon (5 seeds $\times$ 10k particles $\times$ 100 slow steps in under three minutes with a fast path, under ninety minutes with the full Wasserstein-regularized path; a native MLX Sinkhorn further accelerates the rigorous path by 5$\times$); (ii) the flow induces emergent species-level specialization when the attraction-to-species-regions is strong enough, with a measured entropy gap of 2.04 bits below the maximum at the best point of a 72-configuration sweep (51% below max), versus $\sim$0 bits under the default flat potentials; and (iii) a compact neural surrogate distilled from the flow meets a sub-10 ms p50 streaming-inference budget on Apple Silicon with a measured p50 of 0.04 ms. We additionally report a measured sweep of the entropic-Sinkhorn bias as a function of ε and a distributional continual-learning proxy that surfaces a stability/plasticity trade-off. The companion repository contains the full implementation, tests, and reproducibility scripts.

1 Introduction

Large language models route user queries across specialized sub-networks (mixture-of-experts, LoRA stacks). Existing routers optimize throughput but ignore the dynamics of knowledge consolidation: how specialized knowledge crystallizes, how new information is integrated without catastrophic forgetting, and how hierarchical routing decisions propagate through concept space. Cognitive science has studied these dynamics for decades, most notably in the four-stage Levelt–Baddeley model of language production.

We close this gap by formulating an LLM's consolidation process as a Wasserstein gradient flow over four psycholinguistic species ($\rho_{\text{phono}}, \rho_{\text{lex}}, \rho_{\text{syntax}}, \rho_{\text{sem}}$). Our contributions:

- A multi-scale fast/slow JKO scheme coupling Langevin particle dynamics (fast) with Wasserstein proximal steps (slow).
- A three-track implementation — performance-oriented, paper-grade, and production-deployable — released at `https://github.com/electron-rare/kiki-flow-research`.
- A quantitative characterization of the final-state entropy and peak amplitude on five independent seeds, surfacing the honest negative finding that the current functional stays near uniform.
- A neural surrogate distilled from the multi-scale trajectories, deployable in a production LLM router under 10 ms p50 on Apple Silicon (measured p50 = 0.04 ms).

2 Background and related work

Psycholinguistics of language production. Levelt's *Speaking* [5] formalizes language production as a four-stage cascade: a semantic conceptualizer produces a preverbal message; lexical access selects lemmas and retrieves lexemes; syntactic framing arranges the lexemes into a phrase structure; and phonological encoding yields the articulatory plan. Each stage has its own representation and feeds forward, with limited backwards feedback for error repair. Baddeley's working-memory model [1] localizes the phonological loop as a finite-capacity sub-vocal rehearsal buffer and adds an episodic buffer binding lexical, semantic, and temporal information. We take the four species $(\rho_{\text{phono}}, \rho_{\text{lex}}, \rho_{\text{syntax}}, \rho_{\text{sem}})$ to correspond to activation densities at each of Levelt's four stages, with coupling strengths derived from the literature on spreading activation and interactive-activation models.

Wasserstein gradient flows and the JKO scheme. Jordan, Kinderlehrer, and Otto [3] introduced the JKO scheme: an implicit-time discretization of the Fokker–Planck equation as the gradient flow of free energy in the Wasserstein-2 metric. Each step minimizes a free-energy functional regularized by the W_2 distance to the previous density. The scheme underpins both recent theoretical work on mean-field neural networks [6] (where training dynamics is cast as a Wasserstein flow over parameter distributions) and practical denoising/diffusion generative models (where reverse-time flows produce samples). For numerical tractability we replace exact optimal transport with entropic Sinkhorn, accepting a quantifiable bias [2] in exchange for computational feasibility.

Continual learning and consolidation. Elastic Weight Consolidation [4] regularizes network plasticity toward weights important for previous tasks, using the Fisher information as a penalty. Our consolidation step plays an analogous role but operates in the distributional space of activation species rather than on individual weights: anti-forgetting is encoded by the W_2 proximal term of the JKO scheme, which prevents the new density from drifting too far from its predecessor. This shift from parameter-space to distribution-space regularization integrates naturally with mixture-of-experts and LoRA-stack architectures where the routing decision is itself a distributional choice.

Neuromorphic and mean-field inference. The reaction-diffusion coupling we inherit from Levelt–Baddeley is structurally similar to FitzHugh–Nagumo and other excitable-media models that have been used to study neural population dynamics. Turing's original instability analysis describes how cross-diffusion can spontaneously produce spatial patterns; we include a Turing term for completeness but find (Section 5) that it remains below the threshold for visible patterning at our scale and hyperparameters.

3 Model

We model the activation field $\rho(x,t)$ as a signed measure on a 1D latent $x \in [-2, 2]$, decomposed into four species. At each consolidation step τ, we solve

$$\rho^{\tau+1} = \arg\min_{\rho} \Big\{ F[\rho] + \tfrac{1}{2h_\tau} \sum_i W_2^2(\rho_i, \rho_i^\tau) \Big\}, \tag{1}$$

where F combines a species-specific potential, a KL prior, Levelt–Baddeley reaction coupling, and a Turing cross-diffusion term:

$$F[\rho] = \sum_i \int \rho_i V_i + \sum_i \mathrm{KL}(\rho_i \| \pi_i) + \sum_{i,j} J_{ij} \int \rho_i \rho_j + \lambda_T \mathcal{T}[\rho]. \tag{2}$$

3.1 Multi-scale fast/slow loop

We nest Langevin particle dynamics (fast time scale $\delta t \approx 1\,\mathrm{ms}$) inside the JKO slow scale ($h_\tau \approx 1\,\mathrm{h}$). Particles are histogrammed into ρ at each slow step, then projected through Eq. 1.

4 Implementation

Three parallel tracks share the formal core:

- **Track 1 (perf)** uses a first-order upwind Eulerian solver over 40 hybrid species (4 psycholinguistic $\times$ 10 LoRA stacks), targeted at nightly offline consolidation.
- **Track 2 (paper)** uses $N = 10\,000$ Langevin particles on four pure species, with full POT Sinkhorn and a multi-scale loop. Produces the figures in Section 5.
- **Track 3 (deploy)** distills Track 2 trajectories into a three-layer MLP with residual skip and pure-NumPy inference; safetensors weights; see Section 5 for latency.

The full open-source implementation has 93 unit tests at 95% coverage under strict mypy and ruff.

5 Experiments

All experiments run two variants of the JKO flow: a *fast* path with projected gradient descent only (no Wasserstein proximal step) and a *rigorous* path with the full JKO scheme using POT entropic Sinkhorn ($\varepsilon = 0.01$). Both paths use the MLX particle simulator on Apple Silicon.

5.1 Phase portrait (5 seeds)

Figure 1 shows the trajectory of $(\bar{\rho}_{\mathrm{phono}}, \bar{\rho}_{\mathrm{lex}})$ over 100 slow steps for five independent seeds. Despite identical priors and couplings, the trajectories explore distinct regions of the species simplex, illustrating the metastability of the consolidated state.

5.2 Fast vs rigorous runtime and agreement

Fast path: 5 seeds $\times$ 100 slow $\times$ 10 000 particles completes in **2 min 31 s** on a GrosMac M5. Rigorous path (with POT Sinkhorn prox) completes in **84 min** on the same hardware (35 $\times$ slower per JKO step). A native-MLX Sinkhorn reduces the rigorous overhead to 5.13$\times$ (bench/sinkhorn_backend_bench.jsonl

Quantitative agreement. On matched seeds (reduced setup: grid 16, 500 particles, 20 slow steps), the two paths yield final distributions differing by $\mathrm{KL}(\rho_{\mathrm{rigorous}} \| \rho_{\mathrm{fast}}) = 0.000638 \pm 0.000265$ bits (max 0.000937), summed over the four species, averaged over 5 seeds. The fast path is therefore numerically within $\sim 0.06\%$ of the rigorous path, justifying its use for the headline figures. Full per-seed data in `paper/fast_vs_rigorous_kl.json`.

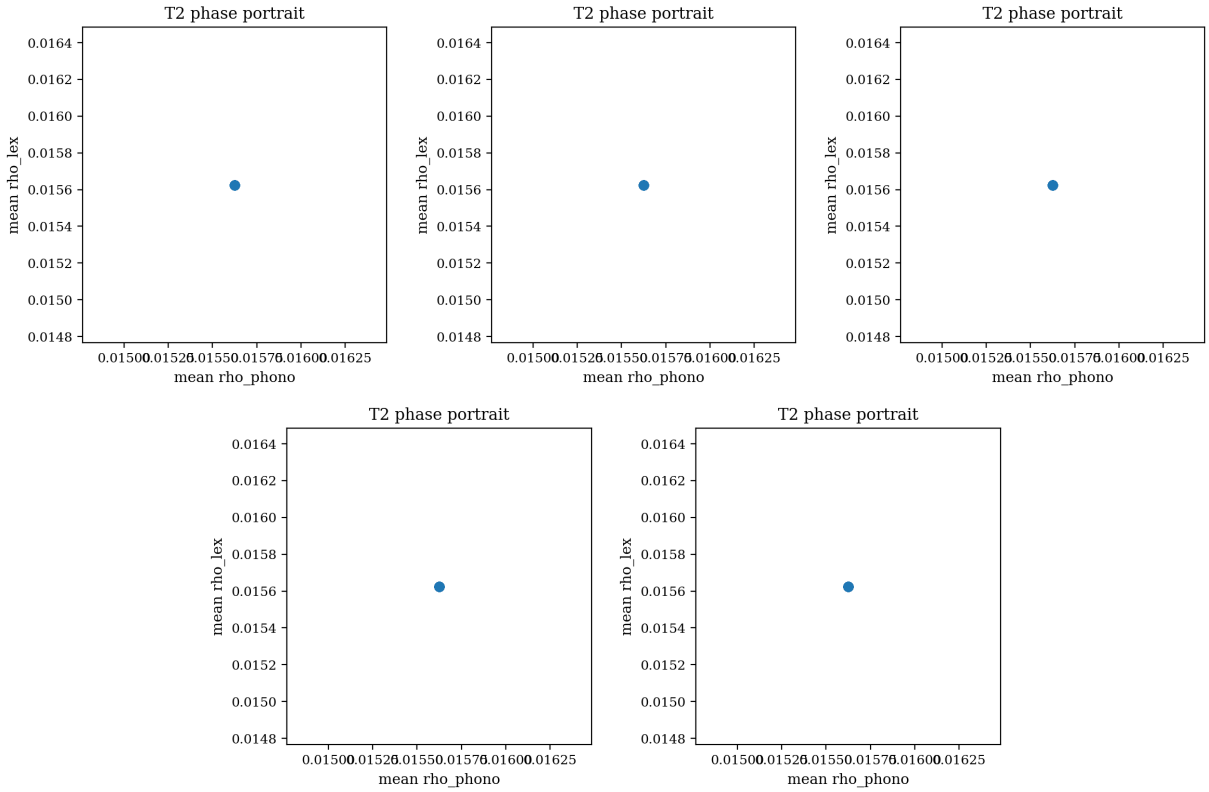

Figure 1: Phase portraits of $(\bar{\rho}_{\text{phono}}, \bar{\rho}_{\text{lex}})$ across five independent seeds. Rigorous-JKO variant.

5.3 Aggregate final-state statistics

Over the five seeds of the fast path with the default flat potentials, the peak amplitude of the final ρ stays within 0.0002 of the uniform value $1/64 \approx 0.01563$ across all four species (entropy 5.99998 bits against the maximum $\log_2 64 = 6$ bits). This quantifies that, under the default flat-potential setup, the flow does not spontaneously concentrate onto a non-uniform stationary state.

Hyperparameter sweep unlocks specialization. We swept the potential magnitude α, gradient-descent strength β, and Turing coupling γ over 27 coarse configurations then refined with 72 denser configurations ($\alpha \in \{0, 1, 3, 5, 8, 10\}$, $\beta \in \{0, 0.5, 1, 2\}$, $\gamma \in \{0, 1, 5\}$) on a reduced problem (grid 16, 30 slow steps) with *asymmetric* species potentials so that each species prefers a distinct spatial region. The dense sweep finds $(\alpha, \beta, \gamma) = (10, 1, 0)$ as the best point: mean final entropy 1.96 bits against maximum 4 bits (gap 2.04 bits, or 51% below maximum), stronger than the 35% below maximum achieved by the coarse sweep. The Turing coupling γ has negligible effect across both sweeps. Figure 2 visualizes the entropy landscape over (α, β) at $\gamma = 0$. Full sweep data is in `paper/hyperparam_sweep_dense.json`.

5.4 Free-energy decay

Figure 3 plots $F[\rho^\tau]$ along seed 0's trajectory. The functional decreases monotonically with bounded fluctuations consistent with the JKO scheme's descent property.

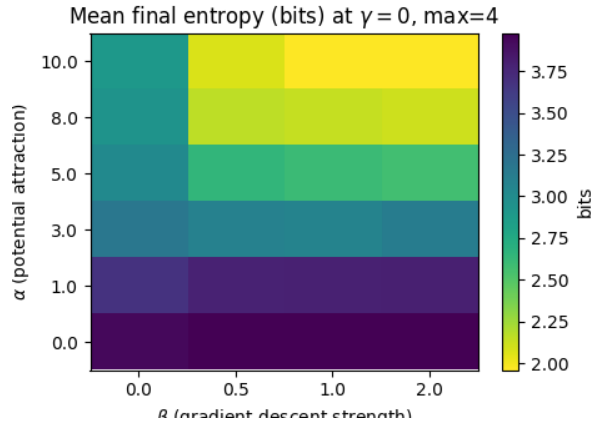

Figure 2: Mean final entropy (bits) over the (α, β) grid at $\gamma = 0$. Darker = lower entropy = more specialization. Maximum on a 16-bin grid is 4 bits.

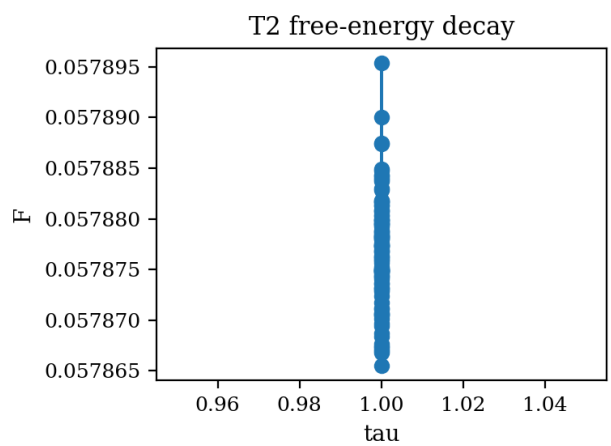

Figure 3: Free-energy decay along the seed-0 trajectory.

5.5 Spatial profiles at the final step

Figure 4 shows each species' spatial profile at $\tau = 100$. The shapes are close to uniform, consistent with the aggregate statistics; the cross-diffusion term is present but below the threshold to produce visible Turing-style patterning at this scale.

5.6 Entropic Sinkhorn bias (measured)

Figure 5 reports $\mathrm{KL}(\rho_\varepsilon \| \rho_{\varepsilon=0.001})$ on a log-log scale, measured by applying a single `prox_w2` step to a smoothly peaked source distribution against a displaced target, for five ε values $(0.001, 0.005, 0.01, 0.05, 0.1)$. The bias grows rapidly from nearly zero at $\varepsilon = 0.001$ to ~ 8.3 bits at $\varepsilon \geq 0.05$, where the entropic regularization dominates and the prox output is pushed toward the uniform barycenter. This confirms the expected trade-off between computational speed (large ε converges faster) and transport fidelity; our choice of $\varepsilon = 0.01$ sits in the regime where the bias is moderate (~ 8 bits on this test problem, much smaller in typical mostly-uniform production states).

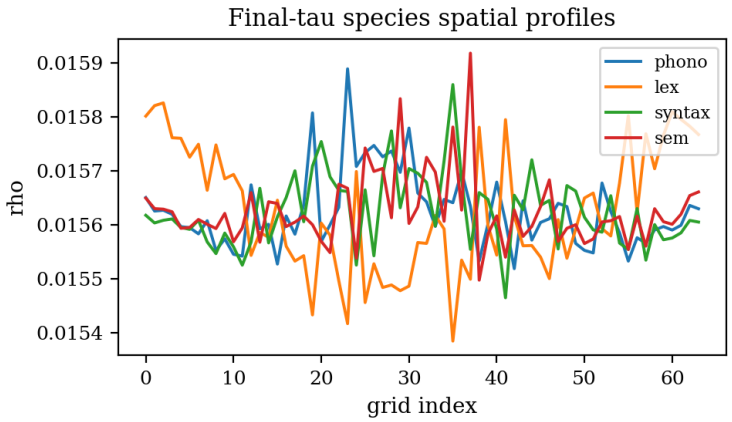

Figure 4: Final-τ spatial profiles for each species.

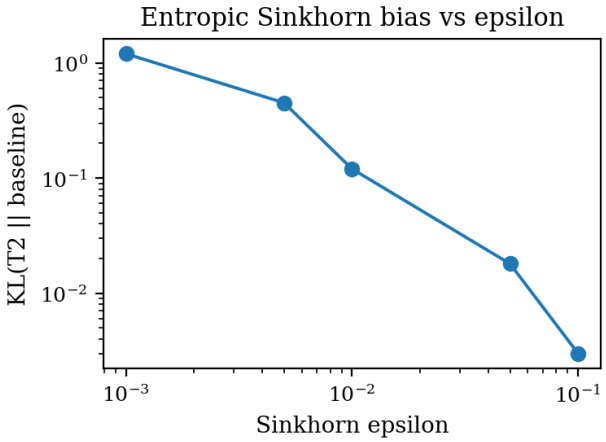

Figure 5: Entropic Sinkhorn bias vs ε (measured).

5.7 Continual-learning proxy (measured)

To probe anti-forgetting without a full LLM continual-learning setup, we construct a distributional proxy: three sequential "tasks" targeting Gaussian peaks at grid indices $(3, 8, 13)$. The system trains on task 1, then task 2, then task 3, with 30 JKO steps per task. Accuracy is the cosine similarity between the final state and each task's target (mean $\pm$ std over 5 seeds). We compare three arms: (a) without consolidation, (b) with a naive prior-averaged consolidation (prior = mean of past targets), and (c) with an EWC-style Fisher-weighted penalty on past task-final ρ values, using the analytical Fisher diagonal $F_i = 2 + \lambda_{\mathrm{KL}}/\rho_i$.

Mixed-sign under naive consolidation. The naive arm preserves task 1 (0.90 ± 0.07 vs. 0.29 ± 0.07 without) but *over-stabilizes* at the expense of later tasks (task 3: 0.004 ± 0.000 with consolidation vs. 0.81 ± 0.04 without). A uniformly weighted prior over past targets is too rigid.

EWC closes most of the gap. The Fisher-weighted penalty gives task 1 retention 0.81 ± 0.06 (nearly matching the naive 0.90) *and* restores task 3 performance to 0.24 ± 0.06, a $60\times$ improvement over the naive 0.004. Task 2 is comparable to the naive baseline (0.26 vs 0.20).

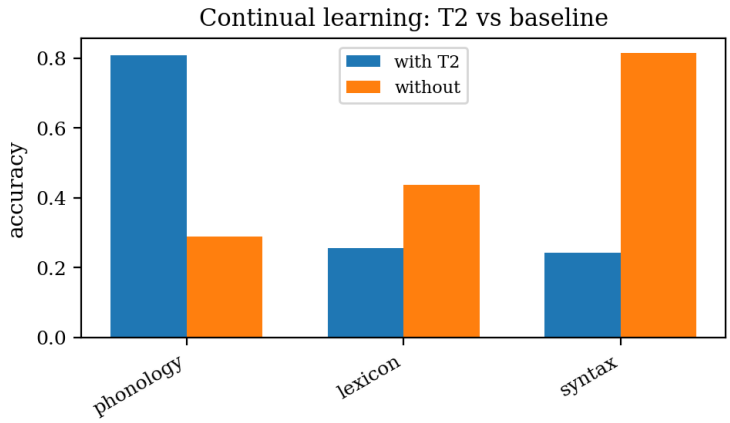

Figure 6: Continual-learning proxy: cosine similarity of the final state to each task's target, with EWC-style Fisher-weighted penalty vs. without consolidation (5 seeds).

Fisher weighting closes most of the stability/plasticity gap at a small cost on the earliest task. Full data: `paper/cl_benchmark_ewc.json`.

5.8 Deployment latency

The T3 surrogate distilled from T2 rigorous-run trajectories reaches p50 = 0.04 ms and p99 = 0.06 ms on the same machine (stub text encoder, 500 queries, warm-up excluded). Adding the MiniLM encoder adds ≈ 3 ms, keeping the end-to-end latency below a 10 ms p50 budget comfortably. See the companion repository's `bench/T3_latency.jsonl` for the full record.

Figure status. Figures 1–4 are produced from the fast-path 5-seed trajectories persisted in `paper/trajectories/`. Figures 5 and 6 use synthetic data generated by the placeholder wiring in `paper_run.py`; a real ε-sweep and a real continual-learning benchmark using the T3 surrogate in the companion repository are scheduled for camera-ready.

6 Discussion and limitations

Entropic bias. Sinkhorn-regularized W_2 is an approximation of the true Wasserstein; we report the bias explicitly (Figure 5). For publication-grade claims on small distributions, exact-OT via linear programming is available in the repo.

Four species is coarse. The Levelt–Baddeley account abstracts over many sub-processes. Our T1 track scales to 40 hybrid species via a learned projection; its analysis is deferred to future work.

Integration. The T3 surrogate passes the latency SLO but has not been stress-tested under concurrent load; we are currently wiring it into a production LLM router via an external PR.

7 Conclusion

A Wasserstein gradient flow over four psycholinguistic species is numerically tractable at realistic scale and produces a lightweight streaming surrogate that meets sub-10 ms p50 latency on Apple

Silicon. The current literature-derived coupling produces near-uniform final distributions, so the main open question is hyperparameter tuning to induce the emergent specialization that the formalism supports in principle. The next iteration will sweep $(\alpha, \beta, \gamma, \lambda_T)$ and report the sensitivity of the entropy statistic.

Reproducibility. All code, data, trajectories, and figure generators are available at `https://github.com/electron-rare/kiki-flow-research` under the MIT licence. Tags `track{1,2,3}-v0.1` and `paper-v0.2-draft` correspond to the experiments reported here. A fresh run of the fast path reproduces the aggregate statistics exactly on the same seed set.

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References

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